



Mini-review

Understanding and modulating exciton dynamics of organic and low-dimensional inorganic materials in photo(electro)catalysis

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ARTICLE INFO

Article history:

Received 29 September 2020

Revised 22 December 2020

Accepted 26 December 2020

Available online 5 January 2021

Keywords:

Excitons

Exciton dynamics

Exciton dissociation

Photo(electro)catalysis

Semiconductors

ABSTRACT

The fundamental study on photo(electro)catalytic reactions is central for the potential application in solving energy and environmental issues. However, the exciton effect, which affects their photo(electro)-catalytic performance especially in organic and low-dimensional inorganic materials, is less explored. Therefore, it is necessary to investigate the exciton dynamics involved in the photo(electro)catalytic process, and the strategies for modulating exciton dynamics are also needed for improving the photo(electro)catalytic performance. In this review, we introduce the exciton dynamics and main exciton characterization techniques, and summarize the mechanisms for modulating exciton dynamics in photo(electro)catalysis. Finally, current challenges and future possible developments for exciton dynamics in the photo(electro)catalytic process are discussed. We hope this review will provide insights on the exciton dynamics in the photo(electro)catalysis, thus inspiring more researches on the materials with strong exciton effect in photo(electro)catalytic applications.

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1. Introduction

Photo(electro)catalytic applications such as water splitting and CO₂ reduction have attracted much attention since they are considered as promising approaches to relieve the energy shortage and environmental pollution issues [1–4]. Among all kinds of photocatalysts, organic (e.g., organic conjugated polymers and small molecules) with the accessibility, chemical versatility and tunable properties and low-dimensional inorganic (i.e., zero-(0D), one-(1D) and two-(2D) dimensional) materials with extremely high specific surface area have been chosen for various photo(electro)catalytic applications [5–13].

After light absorption, the electron and the hole are bound, creating a particle called an exciton. The electron and the hole must be separated to perform the redox reactions in photo(electro)catalysis [14]. Under light irradiation, inorganic bulk photocatalysts generate free charge carriers or loosely bound excitons with binding energies (~0.01 eV) lower than the thermal energy at room temperature ($k_B T$, ~0.025 eV) that can easily dissociate into electrons and holes [10,15]. Thus, the exciton dynamics in such inorganic bulk photocatalysts is not important. In contrast, the stable excitons with relatively high binding energies (typically ranging from 0.25 to 1.2 eV) exist in the organic and low-dimensional inorganic materials upon light excitation [5,15–20], due to the weak intermolecular electronic coupling and low dielectric constant of organic materials, and the reduced dielectric screening in the low-dimensional limit [15,20]. Therefore, their related exciton dynamics should be considered carefully.

The excitons are neutral quasiparticles and they can be classified into singlet and triplet excitons according to their spin multiplicity: singlet exciton (total spin: 0; spin multiplicity: 1); triplet exciton (total spin: 1; spin multiplicity: 3) [19,21]. The excitons can also be sorted based on the strength of Coulomb interaction between the electron and the hole, which is described as the exciton binding energy: the energy difference between the bound state of an electron and a hole (i.e., exciton) and their fully separated state [22]. The Frenkel exciton

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is a tightly bound electron-hole pair with a large binding energy and small Bohr radius (about 1 nm) [23], and the electron is localized in the same or nearby molecule as the hole in the Frenkel exciton [24]. In contrast, the Wannier-Mott exciton is a weakly bound electron-hole pair with a large Bohr radius (about 10 nm) [23], and the electron in the Wannier-Mott exciton is not localized in the vicinity of the hole [24].

In general, three key physiochemical steps are involved in the photo(electro)catalytic charge transfer process: (1) light absorption under irradiation to generate free charge carriers; (2) transport and transfer of free charge carriers to the reaction surface; (3) redox reactions at the material/electrolyte interface [25,26]; and the unwanted free charge recombination may also occur. While for the organic and low-dimensional inorganic materials with strong exciton effect, the dynamics of the excitons, rather than free charge carriers, are involved in the photo(electro)catalytic charge transfer process before step (3), including exciton generation, diffusion, dissociation as well as recombination (Scheme 1). Upon light excitation, electrons from the ground state (S_0) of a material are excited to the singlet excited state and leave positively charged holes in the S_0 , thus producing singlet excitons [18]. The generated singlet exciton can undergo an intersystem crossing (ISC) process and transits to a triplet excited state with the energy lower than that of the singlet excited state, forming a triplet exciton [27,28]. Both singlet and triplet excitons then undergo a diffusion process, followed by dissociating into electrons and holes [12,19,22,29]. The exciton dissociation can happen at the interface with an electron or a hole acceptor, or through interactions with impurities and defect sites, or under the influence of an applied electric field or additional thermal energy that can overcome the exciton binding energy [15,22,30]. The separated electrons and holes then transfer to the surface and participate in the reduction and oxidation reactions, respectively [12,18,29]. At the same time, the exciton recombination (e.g., the radiative decay of the exciton back to the ground state, exciton-exciton annihilation) also occurs and limits the dissociation of the exciton, thus bringing negative effects on the efficiencies of photo(electro)catalytic reactions [22,29]. In addition, the many-body effect (e.g., multiple exciton generation, exciton-exciton annihilation, exciton-plasmon coupling) can also affect the generation, dissociation as well as recombination of the excitons [31,32].

In addition to the above charge transfer, organic and low-dimensional inorganic materials also have exciton-based energy transfer. In detail, energy exchange between excited exciton of catalysts and ground-state reactant is mediated by dipole-dipole interaction or electron exchange interaction in the energy transfer. For example, 1O_2 generation is associated to the energy transfer between ground-state oxygen molecules and triplet excitons in polymeric photocatalysts [27]. Due to the differences in properties like energy level, spin configuration, and charge condition, the energy transfer and charge transfer tend to possess different contributions toward photo(electro)catalysis.

In view of the above descriptions, exciton dynamics, which is important to the organic and low-dimensional inorganic materials with strong exciton effect, is getting more involved in photo(electro)catalysis. But only very few reviews have summarized research works about it in photo(electro)catalysis focusing on either one specific material or modulation strategy so far [5,30]. In this review, we first summarize the characterization techniques of exciton dynamics, then focus on the recent research progress on exciton dynamics (including exciton generation, diffusion, dissociation and recombination) and related modulation strategies in photo(electro)catalysis, and finally a summary and outlook of exciton dynamics in photo(electro)catalysis is presented.

2. Techniques for characterization of exciton dynamics in photo(electro)catalysis

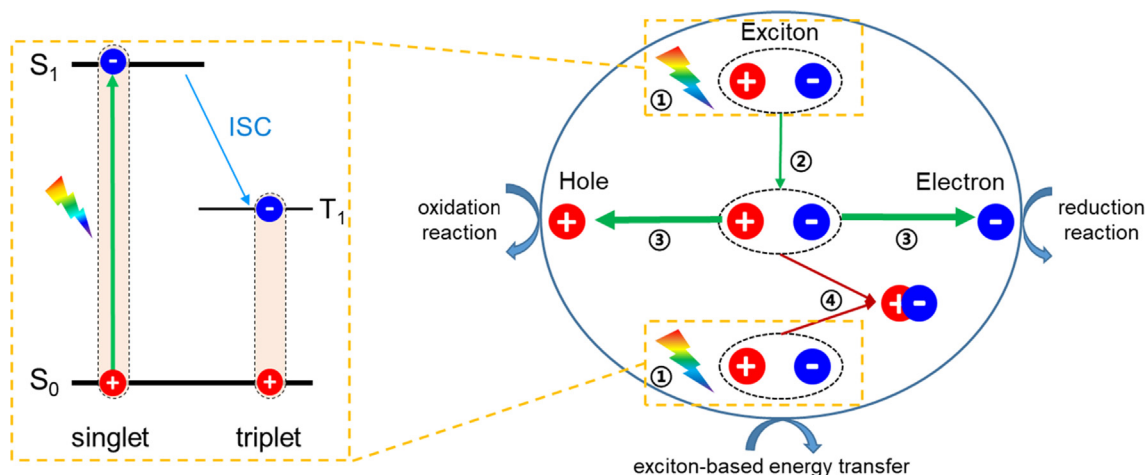
Characterization of excitons using effective techniques is highly important for evaluating the exciton dynamics in photo(electro)catalysis. In this section, mechanisms and applications of techniques for characterization of exciton dynamics in the photo(electro)catalytic process, including transient absorption (TA) spectroscopy, photoluminescence (PL) spectroscopy, (photo)electrochemical measurements and other techniques will be illustrated.

2.1. TA spectroscopy

TA spectroscopy is based on the pump-probe method, including a pump and a probe laser beam with controlled time delay τ (Fig. 1a) [33]. First, the pump beam typically excites 0.1% to tens of percent of the sample within the beam spot. A probe pulse with a relatively weak intensity that avoids multiphoton or multistep processes is then sent through the sample with a tunable τ with respect to the pump pulse [33]. After that, different colors of the transmitted probe beam are separated by a diffraction grating and projected onto a photodiode array. The detected signal is the differential absorption spectrum ΔT between the excited sample ($T(\tau)$) and the unperturbed spectrum T_0 , which can be obtained as a function of τ and wavelength λ by changing the τ between the pump and probe pulses, and it is expressed as ΔT normalized by T_0 ($TA = -\frac{\Delta T}{T_0} = -\frac{T(\tau)-T_0}{T_0}$) [33,37]. If the probe pulse is a monochromatic light, the TA spectrum can be used to follow the kinetics at a specific probe wavelength. In contrast, if the probe pulse is a white-light-continuum, the TA spectrum simultaneously generates kinetics in a broad wavelength range [33,37]. So far, TA spectroscopy has been widely used for recording the ultrafast photoexcitation process, in which information about the exciton dynamics and related mechanisms can be obtained [5], as shown below.

Firstly, the generation of excitons can be recognized by assigning the wavelength or the photon energy from a TA spectrum [36,38–43]. For example, Chowdhury et al. figured out six major exciton-related peaks in the TA spectra of mono-to-quad layer WS_2 flakes, including three bleach features and three positive excited state absorption peaks [38]. Three bleach features correspond to the steady state exciton states, while the positive excited state absorption peaks include two biexcitonic peaks and a peak related to another excited state absorption with a lifetime relatively shorter than the biexcitonic state. Moreover, the phenomenon of multiple excitons produced per absorbed photon is detected by the appearance of the fast multi-exciton decay component in the TA kinetics, when photoexciting above the energy conservation threshold (i.e., larger than twice of the bandgap energy) at low intensity so that each photoexcited sample absorbs at most one photon (Fig. 1b) [34,44–47].

Secondly, after specific peak signal assignment for excitons from the TA spectra, the TA kinetics of specific peaks can be then fitted by various kinetic models (such as the commonly-used multi-exponential functions), from which information such as the time constants of exciton decay, electron transfer, charge separation and recombination is determined on the basis of the fitting parameters [12,35,48–50]. In doing so, the kinetics of excitons in the photo(electro)catalytic process can be obtained. For instance, the time constant representing exci-



Scheme 1. The exciton dynamics in the photo(electro)catalytic process: path 1, exciton generation; path 2, exciton diffusion; path 3, exciton dissociation; path 4, exciton recombination; ISC: intersystem crossing.

ton decay calculated from the TA kinetics in the bismuth oxybromide (BiOBr) nanoplates with oxygen vacancies (OV) is smaller than that of the pristine BiOBr nanoplates, indicating the acceleration of exciton decay in the presence of OV [49].

Thirdly, the charge transfer pathway in the exciton dissociation process can also be revealed from the TA spectrum by assigning the peaks corresponding to the products of charge transfer pathway in the exciton dissociation process [12,35,48,51–54]. Taking the CdSe nanosheet-methyl viologen (MV^{2+}) complex as an example, in which the MV^{2+} is a commonly used electron acceptor and mediator, the dissociation of excitons in this system can be verified through its TA spectrum [35]. Both the TA spectra of CdSe nanosheet- MV^{2+} complex and free CdSe nanosheets show an exciton bleach peak caused by the conduction band (CB) electron state filling induced bleach of exciton band and an exciton absorption peak caused by the exciton transition energy shift in the presence of excitons (Fig. 1c and 1d). The major difference between the spectra of two samples is the gradual formation of an absorption peak representing the MV^+ radical accompanying by the decay of the exciton bleach feature in the TA spectra of CdSe nanosheet- MV^{2+} complex (Fig. 1c), suggesting that excitons generated in the CdSe nanosheets are dissociated by electron transfer to the MV^{2+} and thus forming MV^+ radicals [35].

Other exciton dynamics-related information such as the extent of exciton localization and the exciton–exciton annihilation process can also be reflected in the TA spectrum. The extent of exciton localization can be reflected by the peak position in the TA spectrum. As shown by Guo et al., the red shift of the singlet exciton band in the regioregular poly(3-hexylthiophene) film compared to that of the regiorandom poly(3-hexylthiophene) film indicates that the singlet excitons are more delocalized in the regioregular poly(3-hexylthiophene) film [39]. In addition, the exciton–exciton annihilation process can be identified from the TA spectrum when the decay rate increases with increasing exciton density (Fig. 1e), which is the dominant decay process when the exciton concentration is high [36,39,55–58].

2.2. PL spectroscopy

PL can be divided into prompt fluorescence (PF, radiative decay of singlet excitons) and phosphorescence (PH, radiative decay of triplet excitons) depending on the spin multiplicity of the excited state [27]. Because of its spin-forbidden character, PH usually possesses a much longer lifetime than PF.

Two types of PL spectroscopies are commonly used for characterization: steady-state PL (SS-PL) and time-resolved PL (TR-PL) spectroscopy. The SS-PL spectroscopy measures the PL intensity as a function of emission wavelength; while the TR-PL spectroscopy analyzes the intensity changes of PL spectrum over time.

2.2.1. Steady-state PL spectroscopy

The SS-PL spectroscopy can be used for exciton type assignment, exciton recombination comparison, exciton diffusion length determination and singlet–triplet energy gap calculation.

Firstly, the generation of different types of excitons can also be determined through the SS-PL spectrum [36,40–42,61–64]. For example, from the PL spectrum of monolayer MoS_2 crystals, two kinds of exciton peaks are assigned according to the photon energy as shown by Sun et al., including a dominant emission feature at 1.88 eV representing band-edge (A) exciton and the hot PL peak from B exciton around 2.05 eV [36]. At similar photon energies, Splendiani et al. found two obvious PL emission peaks representing A1 and B1 excitonic transition energies of monolayer MoS_2 flakes with the energy split from valence band (VB) spin-orbital coupling in its PL spectrum (Fig. 2a) [42]. In addition, the defect structure, the number of layers, as well as the size of a material can shift the position or change the width of an exciton peak in a PL spectrum [41,42,62,65].

Secondly, since the SS-PL spectrum originates from the radiative decay of excitons, it can be used to compare the degree of exciton recombination in different samples [27,58,59,66]. For example, cyano-groups ($-C\equiv N$) modified $g-C_3N_4$ (CCN) exhibits significantly reduced PL intensity than that of $g-C_3N_4$, indicating defect engineering can be used to suppress exciton recombination [59]. Furthermore, in general the SS-PL intensity decreases as the temperature increases, which is attributed to the temperature-activated exciton dissociation [67]. Therefore, the exciton binding energy (E_B) can be estimated by fitting the temperature-dependent SS-PL intensity with the Arrhenius equation (Fig. 2c) [60,67].

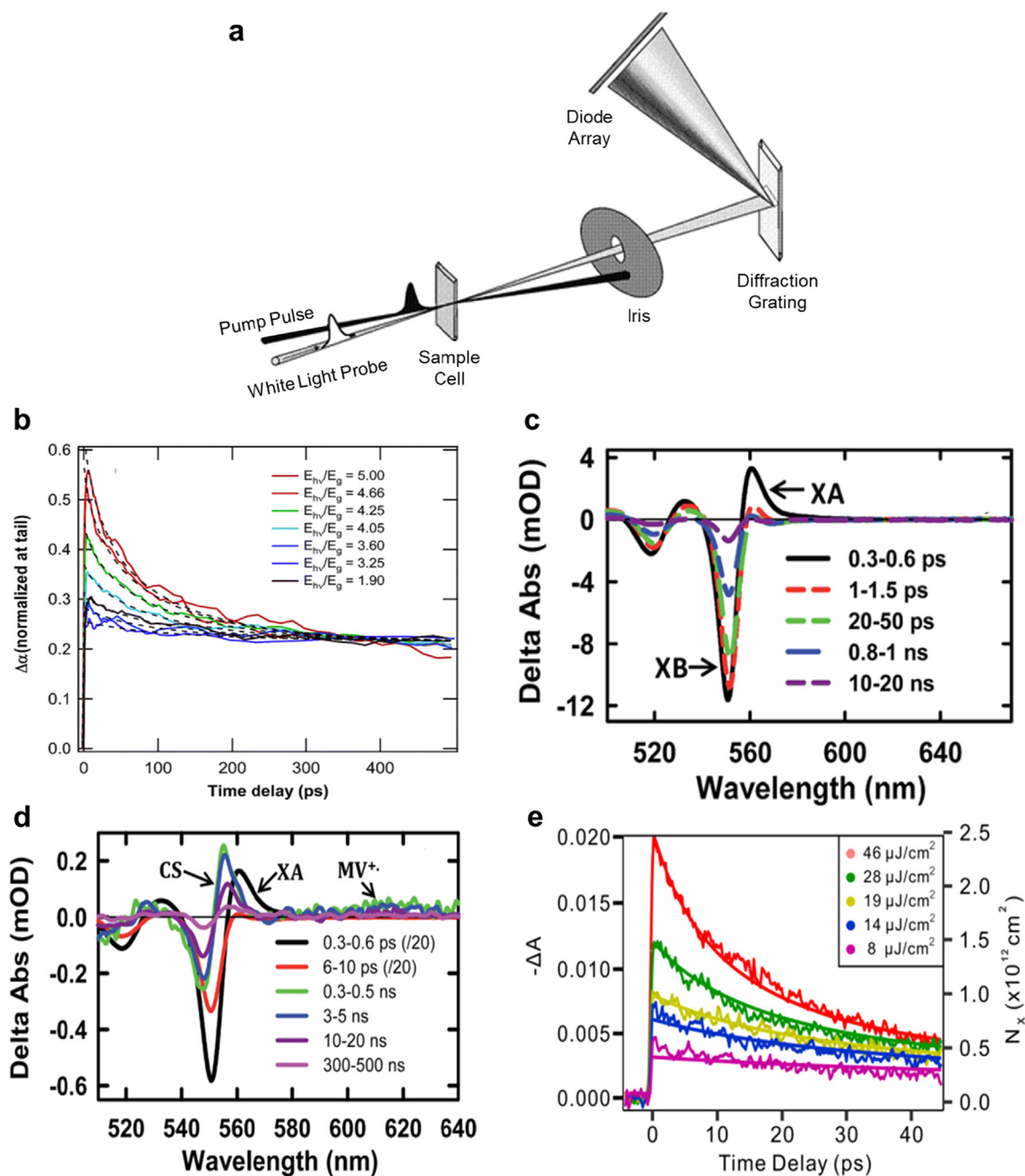


Fig. 1. (a) Schematic depiction of the transient absorption spectroscopy principle. Reproduced with permission [33]. Copyright 2009, Springer Nature. (b) Exciton population decay dynamics of PbSe QDs with a diameter of 5.7 nm obtained by probing intraband (intraexciton) transitions in the mid-IR at different ratios of the energy of the pump photon to that of the bandgap (E_{hv}/E_g). Reproduced with permission [34]. Copyright 2005, American Chemical Society. TA spectra of (c) CdSe nanosheets and (d) CdSe nanosheet-MV²⁺ complex measured with 400 nm excitation (XA, XB and CS in the figures represent exciton absorption, exciton bleach and charge separated state, respectively). Reproduced with permission [35]. Copyright 2015, Royal Society of Chemistry. (e) Exciton dynamics of monolayer MoS₂ at different pump intensities with a time scale of 45 ps. Reproduced with permission [36]. Copyright 2014, American Chemical Society.

Thirdly, the singlet–triplet energy gap, which indicates the efficiency of ISC and triplet exciton generation: the smaller the gap, the higher the efficiency, can also be calculated by measuring the energy difference between the SS-PF and SS-PH peaks [27,58]. For instance, the singlet–triplet energy gap values of the carbonyl functionalized g-CN and the pristine g-CN are calculated to be 0.156 and 0.248 eV, respectively (Fig. 2d), indicating that the incorporation of carbonyl in the g-CN matrix facilitates the ISC process owing to the smaller singlet–triplet energy gap and thus enhances the concentration of triplet exciton [27].

Fourthly, the exciton diffusion length can be determined by measuring its PL intensity either as a function of the added quencher concentration or the layer thickness. The latter method, which is commonly used, measures the PL intensities with and without an exciton quenching layer located adjacent to the material at different thicknesses, and the exciton diffusion length is then obtained after modelling the PL curve with a diffusion equation [68–72].

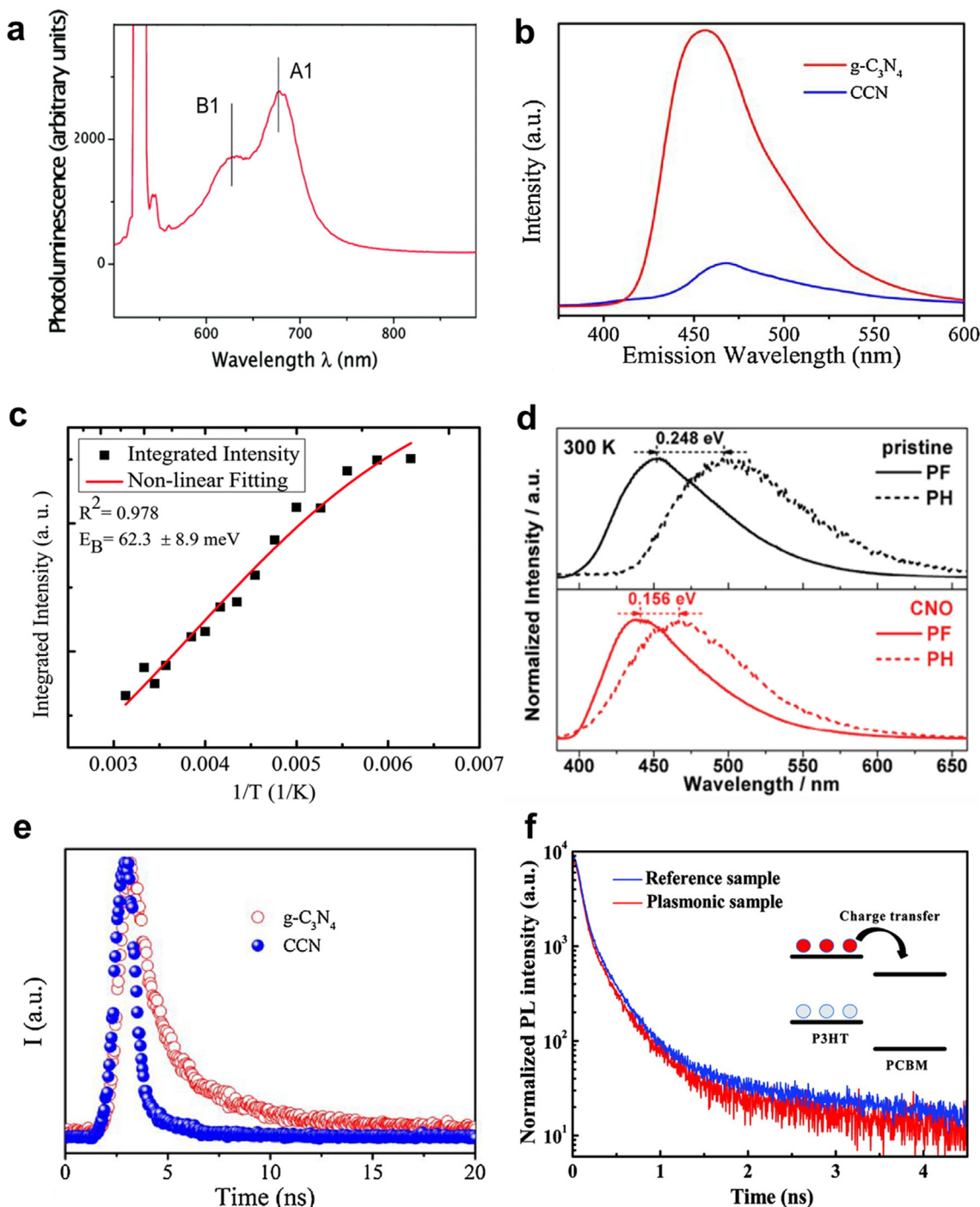


Fig. 2. (a) PL spectrum of monolayer MoS₂ flake. Reproduced with permission [42]. Copyright 2010, American Chemical Society. (b) Steady-state PL spectra of g-C₃N₄ and CCN. Reproduced with permission [59]. Copyright 2018, Elsevier B.V.. (c) Temperature dependent data of integrated SS-PL intensity of the CH₃NH₃PbI_{3-x}Cl_x film. The solid line is the fitting. Reproduced with permission [60]. Copyright 2014, Royal Society of Chemistry. (d) Normalized SS-PL and SS-PH spectra of pristine and carbonyl functionalized g-CN (PF and PH in this Figure represent fluorescence and phosphorescence, respectively). Reproduced with permission [27]. Copyright 2016, WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (e) TR-PL spectra of g-C₃N₄ and CCN. Reproduced with permission [59]. Copyright 2018, Elsevier B.V.. (f) PL decay profiles for the P3HT/PCBM blends in the reference and plasmonic samples. Excitation source: 470 nm pulsed laser. Inset: Schematic representation of the charge transfer process in the P3HT/PCBM blend. Reproduced with permission [77]. Copyright 2011, American Chemical Society.

2.2.2. Time-resolved PL spectroscopy

The TR-PL spectrum and its curve-fitting result can be used to calculate the exciton lifetime of a sample, investigating the associated exciton recombination (radiative decay) and dissociation processes [27,41,58,59,66,73–75]. Furthermore, the coefficient of each lifetime reflects the exciton decay fraction in the corresponding pathway [76], thus helping to understand and compare the contribution of different exciton decay processes in the photo(electro)catalytic reaction. For example, if the TR-PL spectrum is well fitted with the bi-exponential function $y = A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2}$, it reveals that excitons decay through two pathways, where τ_1 and τ_2 are two decay components while A_1 and A_2 represent the fractions of decaying in these two pathways.

As reported, the emission decay curves of g-C₃N₄ and CCN are fitted using the bi-exponential kinetics function [59]. Two decay components of τ_1 and τ_2 are derived: the shorter lifetime component of τ_1 is originated from the nonradiative quenching, whereas the longer one of τ_2 represents the recombination of free excitons in the material. As a result, compared to the pristine g-C₃N₄ ($\tau_1 = 1.35$ ns, $\tau_2 = 6.37$ ns, average PL lifetime of 3.9 ns), CCN shows both shorter lifetimes of nonradiative quenching and the recombination of excitons ($\tau_1 = 0.49$ ns, $\tau_2 = 3.554$ ns, average PL lifetime of 1.2 ns), indicating the reduced exciton recombination and promoted exciton dissociation in CCN (Fig. 2e) [59].

In addition, TR-PL spectrum can also be used to probe how the plasmonic effects influenced the exciton dynamics [77]. As shown in Fig. 2f, the reference sample has a biexponential fitting with average PL lifetime of 0.36 ns; while the plasmonic sample (with Au nanoparticles) has a triexponential fitting with average PL lifetime of 0.27 ns. The decreased exciton lifetime and one more decay component in plasmonic sample come from the plasmon-exciton coupling, which facilitates the charge transfer process for exciton dissociation [77].

2.3. Other techniques

Other techniques, including ultraviolet–visible (UV–vis) absorption spectra, Raman spectroscopy and theoretical calculation, can be regarded as the additional support of TA and PL spectroscopy for exciton dynamics characterization. Firstly, UV–vis absorption spectra can be used to explore the exciton generation [49,78]. Secondly, Raman spectroscopy can help to probe the exciton-photon interaction and exciton-plasmon coupling by the peak assignment and peak intensity comparison [79–81]. Thirdly, the theoretical calculation can be used to find out the possible ISC channels from singlet excitons to triplet excitons [82], and give some insights into the band modifications via structure engineering (such as defects, vacancies, dopants, distortion and thickness), which is significant for exciton dynamics [5,49,74].

Combining the UV–vis absorption spectra and the theoretical calculation, Tian et al. found that the heptazine-based melon with $\text{—C}\equiv\text{N}$ defects (HM-d) promotes the exciton generation compared to the pristine HM [83]. On one hand, UV–vis absorption spectra shows that the absorption edge of HM-d has a progressive redshift with the increasing defect concentration; on the other hand, according to the theoretical calculation, HM-d has larger oscillator strength values between S_0 and excited states owing to its higher transition dipole moment originated from the increased HOMO–LUMO overlap as well as the smaller energy differences between S_0 and excited states. Both techniques indicate larger light absorption and more singlet excitons are obtained in HM-d. Another example shows that the plasmon-exciton coupling for chemical reactions can be measured by surface enhanced Raman spectroscopy [81]. Compared with g-C₃N₄/Ag, C₃N₅/Ag has a significantly decreased reactant peak intensity and an obviously increased product peak intensity at a lower laser power, indicating its promoted surface-plasmon-to-exciton conversion efficiency as well as accelerated chemical reaction [81].

3. Exciton dynamics in photo(electro)catalysis and related modulation strategies

The exciton dynamics is important to the photo(electro)catalytic performance of semiconductors with a strong exciton effect, so it is necessary to understand and modulate the exciton dynamics in the whole process. In this section, exciton dynamics, including exciton generation, diffusion, dissociation and recombination in the photo(electro)catalysis, and the related modulation strategies will be discussed.

3.1. Exciton generation

The generation of excitons is seen as the prerequisite for the photo(electro)catalytic reaction in the organic and low-dimensional inorganic materials. In the following parts, strategies of surface plasmon resonance (SPR), multiple exciton generation and defect engineering for promoting exciton generation will be introduced in detail.

3.1.1. Surface plasmon resonance

In the semiconductor/plasmonic metal structure, the SPR effect, which is the coherent collective oscillation of conduction electrons surrounding the plasmonic metal surfaces (e.g., Au, Ag) activated by incident light, can enhance the exciton generation in the semiconductor based on the light scattering mechanism as well as the near-field mechanism [77,86,87]. In detail, on one hand, the presence of plasmonic metal nanoparticles can enhance light scattering, increasing the average photon path length in semiconductor/plasmonic metal structure and thereby promoting light absorption and exciton generation [87]. On the other hand, the SPR effect can enhance near-field intensity of the photoexcited plasmonic metals, resulting in promoted generation of excitons in the semiconductor nearby because the rate of exciton generation is proportional to the square of near-field intensity [87].

Various examples have confirmed that the SPR effect can promote exciton generation for improved photo(electro)catalytic performance [78,87,88]. For instance, compared to the pristine MoS₂, the Au@MoS₂ hybrid nanostructure, where the Au nanoparticles are integrated with MoS₂, produces strong near-field enhancement and then drastically promotes the exciton generation of MoS₂ (Fig. 3a) [78]. Furthermore, as the amount of Au increases from monomer to multimer (monomer means only 1 nanoparticle while multimer means 2–3 nanoparticles), the number of generated excitons in the Au@MoS₂ hybrid nanostructure increase (Fig. 3a and 3b) owing to the strongly in-phase coupling between Au nanoparticles, which further enhances the intensity of the near-field under the on-resonant excitation light (i.e. 532 nm for the Au monomer and the Au multimer; 630 nm for the Au multimer) [78].

3.1.2. Multiple exciton generation

In addition to the SPR, the multiple exciton generation can also appear during the light absorption process, promoting the generation of excitons [84]. The multiple exciton generation, which is also known as the impact ionization or carrier multiplication, is a process by which all or some of the excess energy can be used for generating two or more excitons from a single absorbed photon rather than loses from the system as heat when the photon energy is greater than the bandgap of a semiconductor (Fig. 3c and 3d) [44,46,89–92]. The multiple exciton generation effect can be enhanced in semiconductors with nanostructures such as semiconductor QDs and nanocrystals compared to bulk semiconductors due to the quantum confinement effect, which enhances the Coulomb interaction and reduces the density of final states

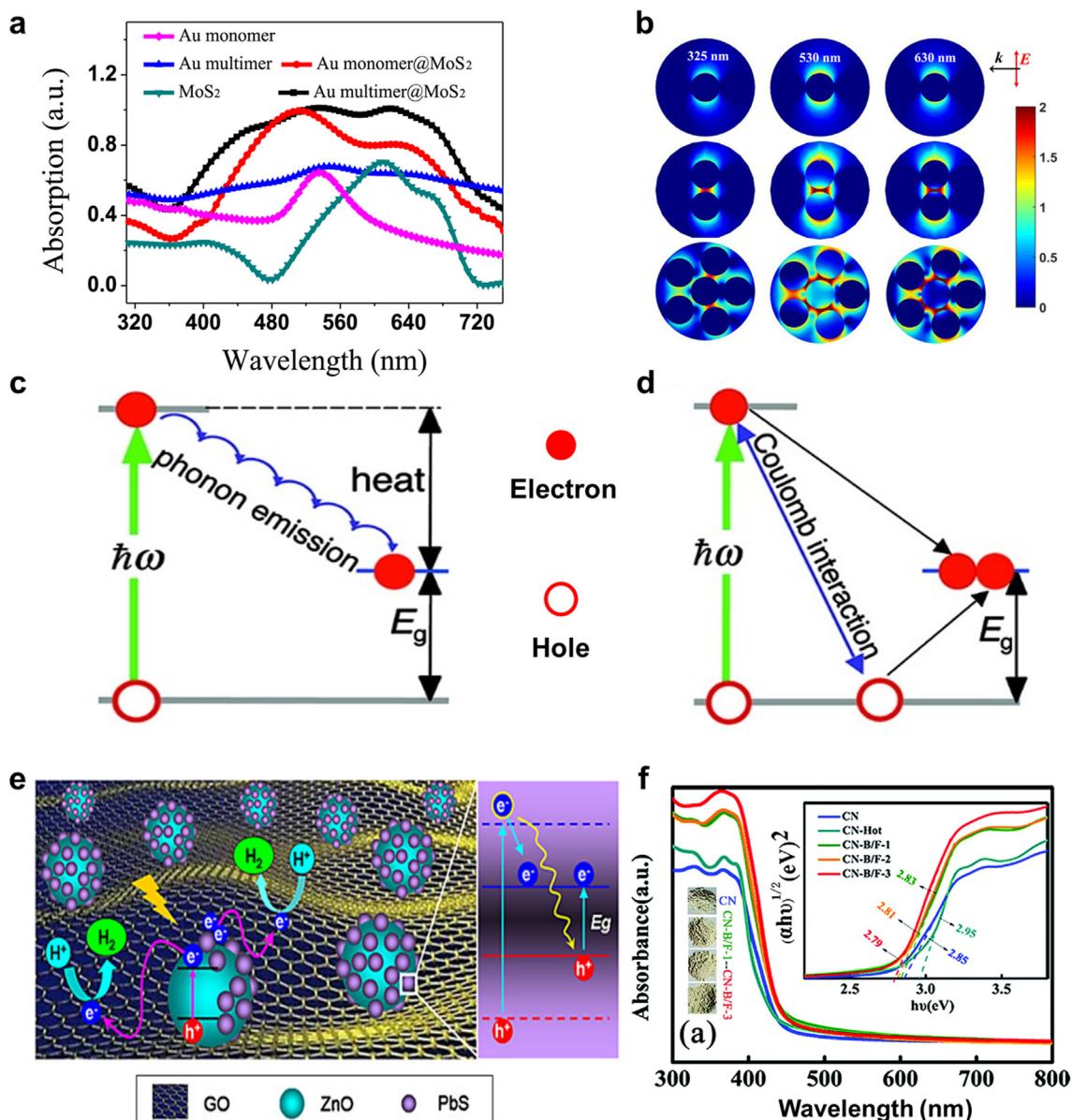


Fig. 3. (a) UV–visible absorption spectra of the Au monomers, Au multimers, MoS₂ spheres, Au monomer@MoS₂, and Au multimer@MoS₂. Reproduced with permission [78]. Copyright 2016, Elsevier Ltd.. (b) The top view of the electrical field distribution of the Au monomer@MoS₂, Au dimer@MoS₂, and Au multimer@MoS₂ under different excited wavelengths. Reproduced with permission [78]. Copyright 2016, Elsevier Ltd.. (c) Absorption of a single photon (green arrow) produces an exciton. In this case, the photon energy in excess of the energy gap is dissipated as heat by exciting phonons (blue arrows). Reproduced with permission [46]. Copyright 2008, American Chemical Society. (d) Multiple exciton generation process is a process in which absorption of a single photon produces more than one electron-hole pairs. It shows that a high-energy CB electron excited by a photon loses its excess energy by transferring it via the Coulomb interaction to another VB hole, rather than being dissipated as heat (like Fig. 3c). As a result, this hole is excited across the energy gap to produce a secondary exciton. Reproduced with permission [46]. Copyright 2008, American Chemical Society. (e) A schematic diagram of the structure of the ZnO@PbS/GO photocatalyst and the electron transfer mechanism in the system. Reproduced with permission [84]. Copyright 2014, Elsevier B.V.. (f) UV–vis absorption spectra and band gaps of CN, CN-hot, CN-B/F-1, CN-B/F-2 and CN-B/F-3. Reproduced with permission [85]. Copyright 2019, Royal Society of Chemistry.

because of a discretization of the electronic states, and thus enhances the multiple exciton generation process and reduces the heat lost [44,46,47,91–93]. In order to have a positive effect on the photo(electro)catalytic activity, the produced multiple excitons must be separated prior to exciton–exciton annihilation, which may decrease the number of excitons [44]. Therefore, the structure with the multiple exciton generation for the photo(electro)catalytic application needs to be well designed to make it effective.

Studies have shown that multiple exciton generation can improve the photo(electro)catalytic efficiency [84,94,95]. Shi et al. first introduced the multiple exciton generation property into the semiconductor QDs for photocatalytic H₂ evolution from water [84]. The H₂ generation rate of the ZnO@PbS/GO, in which the PbS QDs is used as the multiple exciton donor, the ZnO serves as the conduction bridge of the photoelectrons generated from PbS QDs, and the GO is considered as the electron acceptor and the active site for H₂ production (Fig. 3e), is much higher than that of the ZnO/GO, owing to the multiple exciton generation from the PbS QDs of the ZnO@PbS/GO sample.

3.1.3. Defect engineering

Defect engineering, including vacancies, dopants and lattice disorders, is widely used in photo(electro)catalysis to extend the light absorption range and promote exciton generation through narrowing the bandgap [30,85]. Generally, the weaker bonding at defect sites

reduces splitting between bonding and antibonding orbitals compared with valence and conduction band states and thus gives rise to electronic states within the bandgap. The electronic states narrow the bandgap or act as midgap states and thus extend the light absorption range for enhanced exciton generation [30]. For example, compared to the pure CN, there is a redshift in the absorption edge after the incorporation of B and F elements into CN structure (CN-B/F), and the corresponding bandgap decreases from 2.85 to 2.79 eV with the increased doping concentration, which benefits for exciton generation (Fig. 3f) [85].

3.2. Exciton diffusion

The photogenerated excitons then diffuse to the defect sites or interfaces of the heterojunction where they dissociate into electrons and holes. The average distance that between exciton generation and recombination is called the exciton diffusion length, and dissociation of an exciton needs to happen within this distance [22,69]. However, an exciton can only diffuse a short distance (e.g., the exciton diffusion length is less than 10 nm for many conjugated polymers) during their lifetime, so it is necessary to increase the exciton diffusion length and/or adjust the material size similar to the order of the exciton diffusion length, thus increasing the probability of exciton dissociation [71,96]. In this section, enhancing the degree of crystallinity and building nanostructures are discussed to solve the exciton diffusion length issue.

Enhancing the degree of crystallinity can increase the exciton diffusion length owing to the reduction of non-radiative losses at grain boundaries, hence benefiting for the exciton dissociation or exciton-involved energy transfer process [19,97]. For example, Sim et al. found the exciton diffusion length in poly(3-hexylthiophene) (P3HT) is increased from 3 to 7 nm as the degree of crystallinity increased by a factor of 6 [98]. The increase in P3HT crystalline order can improve the spectral overlap integral between the donor fluorescence and acceptor absorption and reduce the distance between chromophores, thereby enhancing Förster-mediated exciton diffusion [98].

Furthermore, building nanostructures can reduce the domain size of the materials similar to the exciton diffusion length (~nm), increasing the probability of exciton dissociation [71,96]. For instance, compared to the bulk g-CN, the size of g-CN nanorod array synthesized by Gong et al. is closer to the exciton diffusion length to suppress more exciton recombination, resulting in an excellent photocurrent density of 120.5 $\mu\text{A cm}^{-2}$ at 1.23 V versus reversible hydrogen electrode under one sun irradiation [99].

3.3. Exciton dissociation and recombination

The nature of exciton dissociation can be seen as to overcome the binding energy of the exciton, which is caused by the Coulomb attraction between the electron and the hole involved in the exciton [22]. Since the free electrons and holes generated from the exciton dissociation directly participate in the charge-involved photo(electro)catalytic reactions, it is the key step in the whole exciton dynamics which needs to be focused on.

Similar to the reverse process of exciton dissociation, two major exciton recombination pathways can be identified in the photo(electro)catalytic process. One is the radiative decay of excitons back to the ground state, in which the electrons in the excited state go back to the ground state and recombine with the holes [29,32]. The other is called the exciton–exciton annihilation, also known as the Auger recombination [32,55,100]. It can be divided into two categories: singlet and triplet exciton–exciton annihilation. For singlet exciton–exciton annihilation, two singlet excitons are involved in the exciton–exciton annihilation process, in which one singlet exciton relaxes to the ground state and the other is promoted to a higher singlet excited state. The singlet exciton at higher energy state can relax back to lower energy state with excess energy dissipated in the form of phonons [19,31,57,101]. Unlike singlet exciton–exciton annihilation, the triplet exciton–exciton annihilation (which is also called triplet–triplet annihilation) process refers to an energy transfer mechanism between two excitons in their triplet states, in which two triplet excitons collide to produce one singlet exciton [58]. It can be concluded that radiative decay reduces the overall number of excitons, thereby reducing the performance of photo(electro)catalysis; while single (or triplet) exciton–exciton annihilation reduces the number of single (or triplet) excitons, leading to decreased singlet (or triplet) exciton-involved performance.

In the following section, in order to enhance the efficiency of photo(electro)catalytic reactions, strategies for promoting exciton dissociation and suppressing exciton recombination, including defect engineering and hetero/homojunction constructing will be summarized.

3.3.1. Defect engineering

The defects work as electron or hole accepting sites in a material [103], where the electrons or holes in excitons migrate to the defects owing to the electrostatic attraction between them, weakening the interaction inside the exciton and promoting the dissociation of excitons into the free electrons and holes to participate in photo(electro)catalytic reactions.

Studies have shown that introduction of different functional groups in g-CN is considered as an effective way to create defects for promoting exciton dissociation [59,66,99,104–106]. It has been found that the amine (–NH–) groups located at the edges of g-CN serve as hole-accepting sites to interact with the electrons in excitons for enhancing exciton dissociation since the N atoms in amine groups have lone-pair electrons (Fig. 4a) [99]. The dissociated electrons participate in the delocalized π -bands of the g-CN, providing better electron mobility and enhanced photoelectrochemical performance, while the dissociated holes go through the O_2 evolution reaction. Besides the –NH– groups, the $\text{—C}\equiv\text{N}$ groups in the HM-d are also seen as the dissociation sites for excitons, and the degree of exciton dissociation depends on the $\text{—C}\equiv\text{N}$ concentration (Fig. 4b) [83]. With increasing $\text{—C}\equiv\text{N}$ concentration, the defect-mediated-trap state promotes exciton dissociation owing to the built-in electric field between $\text{—C}\equiv\text{N}$ and neighbored –NH– groups with different electronegativities until reaching to the highest electron concentration as well as the photocatalytic hydrogen evolution reaction (HER) activity. However, excess $\text{—C}\equiv\text{N}$ groups act as recombination centers, leading to severe exciton recombination and resulting in decreased exciton dissociation as well as HER activity [83].

Moreover, the vacancy existed in a material is also considered as the defect for exciton dissociation [49,102,107,108]. For example, the OV introduced in the BiOBr nanoplates are responsible for the dissociation of excitons in the system, thus providing better performance in charge-carrier-involved photocatalytic reactions such as superoxide radical generation and selective oxidative-coupling reactions [49]. The density functional theory (DFT) calculation shows that the introducing OV can distort the surrounding localization of VB and CB edge states, leading to the dissociation of excitons. Compared with the excitonic pathway in the pristine BiOBr nanoplates, which exhibits a two-step

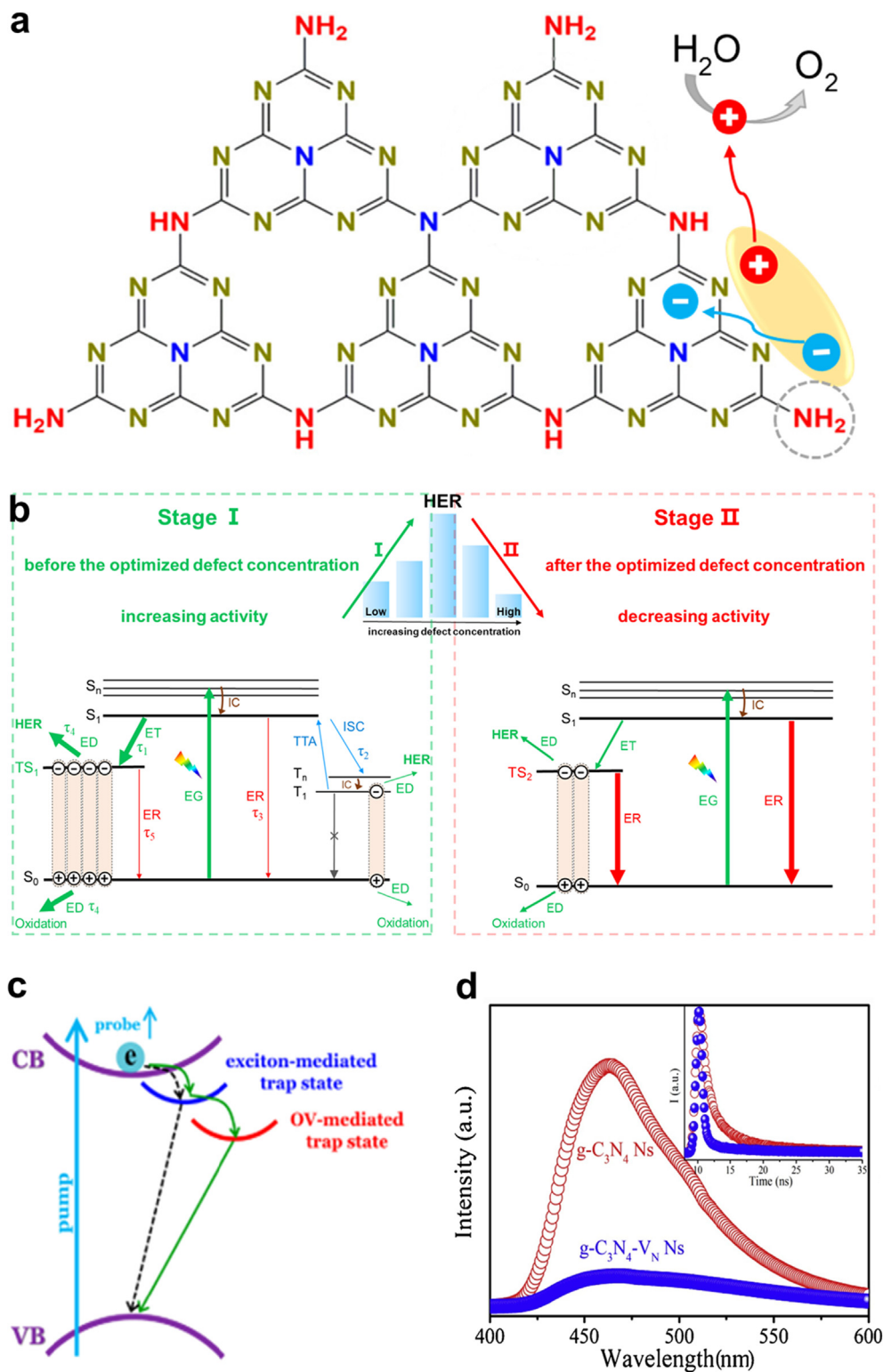


Fig. 4. (a) Schematic diagram of exciton dissociation at the amine groups located at the edges of the g-CN nanorod. Reproduced with permission [99]. Copyright 2018, American Chemical Society. (b) Schematic illustration of the defect concentration-dependent exciton dynamics involved in the photocatalytic HER. IC: internal conversion; ISC: intersystem crossing; TTA: triplet-triplet annihilation; EG: exciton generation; ET: exciton transfer; ED: exciton dissociation; ER: exciton recombination; TS: trap state (TS_1 : exciton and low concentration defect-mediated trap state; TS_2 : exciton and high concentration defect-mediated trap state); the cross in gray: no phosphorescence. Reproduced with permission [83]. Copyright 2020, American Chemical Society. (c) A schematic diagram of photophysical processes involving exciton- and OV-mediated trap states in the BiOBr nanoplates. Reproduced with permission [49]. Copyright 2018, American Chemical Society. (d) Schematic diagram of SS-PL and TR-PL spectra (insert) of $\text{g-C}_3\text{N}_4$ Ns and $\text{g-C}_3\text{N}_4\text{-V}_\text{N}$ Ns. Reproduced with permission [102]. Copyright 2020, Elsevier B.V..

pathway with the excitonic state as an intermediate trap state (black dotted lines in Fig. 4c), a three-step relaxation pathway deduced from TA results of the BiOBr with OV reveals that electrons undergo a fast relaxation from the CB minimum to the exciton-mediated trap state after the photoexcitation, after which the resulting excitons relax to the OV-mediated trap state and dissociate into electrons and holes (green solid lines in Fig. 4c) for charge-carrier-involved photocatalytic reactions such as superoxide generation and selective organic synthesis [49].

Similar to the exciton dissociation mechanism, the introduced defects can also be used as the sites for temporarily trapping the electrons or holes involved in the excitons, thus suppressing exciton recombination [30]. As shown in Fig. 4d, compared to the pristine graphitic carbon nitride nanosheet (g-C₃N₄ Ns), nitrogen vacancy modified g-C₃N₄ nanosheet (g-C₃N₄-V_N Ns) has significantly reduced SS-PL intensity as well as lower exciton lifetime from TR-PL, which originates from the exciton recombination, indicating the introduction of nitrogen vacancy can suppress exciton recombination [102].

3.3.2. Hetero/homojunction constructing

Apart from the defect structure, excitons can also dissociate at the interface of a hetero/homojunction owing to its built-in electric field, which can drive the opposite migration of the electrons and holes involved in the excitons, suppressing exciton recombination and promoting the photo(electro)catalytic reactions [35,52,74,110–114]. For example, 2D-2D p-MoS₂/n-MgIn₂S₄ heterojunction has an internal electric field constructed within the space charge region owing to their type II staggered band alignment, which inhibits the fast recombination of excitons and promotes exciton dissociation (Fig. 5a and 5b) [109]. As a result, there is a quick migration of electrons that takes place from more negative potential of p-MoS₂ to the less negative potential of n-MgIn₂S₄ and holes migrate to the opposite direction after light excitation.

Homojunction has the similar function as heterojunction in promoting exciton dissociation and suppressing exciton recombination [74,115]. Wang et al. proposed that both singlet and triplet excitons dissociated into electrons and holes at the order-disorder interfaces of SC-HM, which can be seen as a homojunction-like structure [74]. DFT simulations demonstrate that the ordered g-CN chain possesses lower HOMO and LUMO energy levels compared with the disordered one, which promotes the electron injection toward the ordered chains and hole blocking in the disordered chains, resulting in selective extraction of electrons from the SC-HM and suppression of charge recombination (Fig. 5c and 5d) [74]. Meanwhile, the conjugated structure of ordered chains favors electron transport, accelerating the migration of electrons to the surface for reactions.

4. Summary and outlook

In this minireview, the recent research progress about exciton characterization, exciton dynamics as well as its modulation in the photo(electro)catalysis has been summarized. However, there are still some remaining issues need to be solved. Hence, in this section, we will describe some of the current challenges and future perspectives on the exciton dynamics in the photo(electro)catalytic process.

- (1) *Developing exciton characterization techniques*: Currently, TA and PL are two widely used techniques for characterization of exciton dynamics in photo(electro)catalysis. However, studies have shown that they also have some drawbacks. For example, the curve-fitting of TA and PL spectra is relative complicated since various assumptions and fitting conditions are required [36,116]. Furthermore, it is also difficult to distinguish whether the population of excitons obtained from the TA spectrum represents all the number or part of them participated in the whole photo(electro)catalytic reaction [116]. To tackle these problems, a combination of real-time and *in situ* experimental characterization techniques for exciton dynamics under the photo(electro)catalytic reaction conditions as well as theoretical calculations based on real-time time-dependent DFT can be considered, in order to acquire more comprehensive and detailed information about photo(electro)catalysis-related exciton dynamics.
- (2) *Exploring the mechanism of energy-transfer-dominated photo(electro)catalysis in depth*: Charge transfer and energy transfer coexist in organic and low-dimensional inorganic materials. As compared to the charge-transfer-dominated reaction, the energy-transfer-dominated reaction is not constrained by the redox properties of the reactants; instead, it mainly depends on the triplet state energies of the reactant and the excited photo(electro)catalyst [117]. As a result, the reactions, which are unlikely via charge transfer, occur under the exciton-based energy transfer mechanism, increasing the diversity of photo(electro)catalytic reactions. However, unlike the well-studied charge transfer mechanism, the exciton-based energy transfer mechanism is still in its infancy for organic and low-dimensional inorganic materials due to their complicated coupling relationships (such as spin-orbit coupling and dipole-dipole coupling) [118,119]. To address this problem, developing feasible modulation methods (like structural modification or co-catalyst design) as well as precise experimental/theoretical characterizations of the coupling-dominated exciton dynamics is necessary, which will open up rich possibilities of solar energy utilization in photo(electro)catalysis.

In a word, this minireview focuses on the exciton dynamics of organic and low-dimensional inorganic materials with strong exciton effect in photo(electro)catalysis, including its basic concept, characterization and modulation strategies. Furthermore, the possible challenges to further improve photo(electro)catalytic performance are pointed out. The modulation strategies of exciton dynamics (including generation, diffusion, dissociation and recombination of excitons) summarized in this minireview offer helpful guidance and reference to the future rational design and construction of materials with strong exciton effect for photo(electro)catalysis. With ongoing progress of science and technologies, we believe that thorough comprehension of exciton dynamics will be beneficial for in-depth understanding of photo(electro)catalysis.

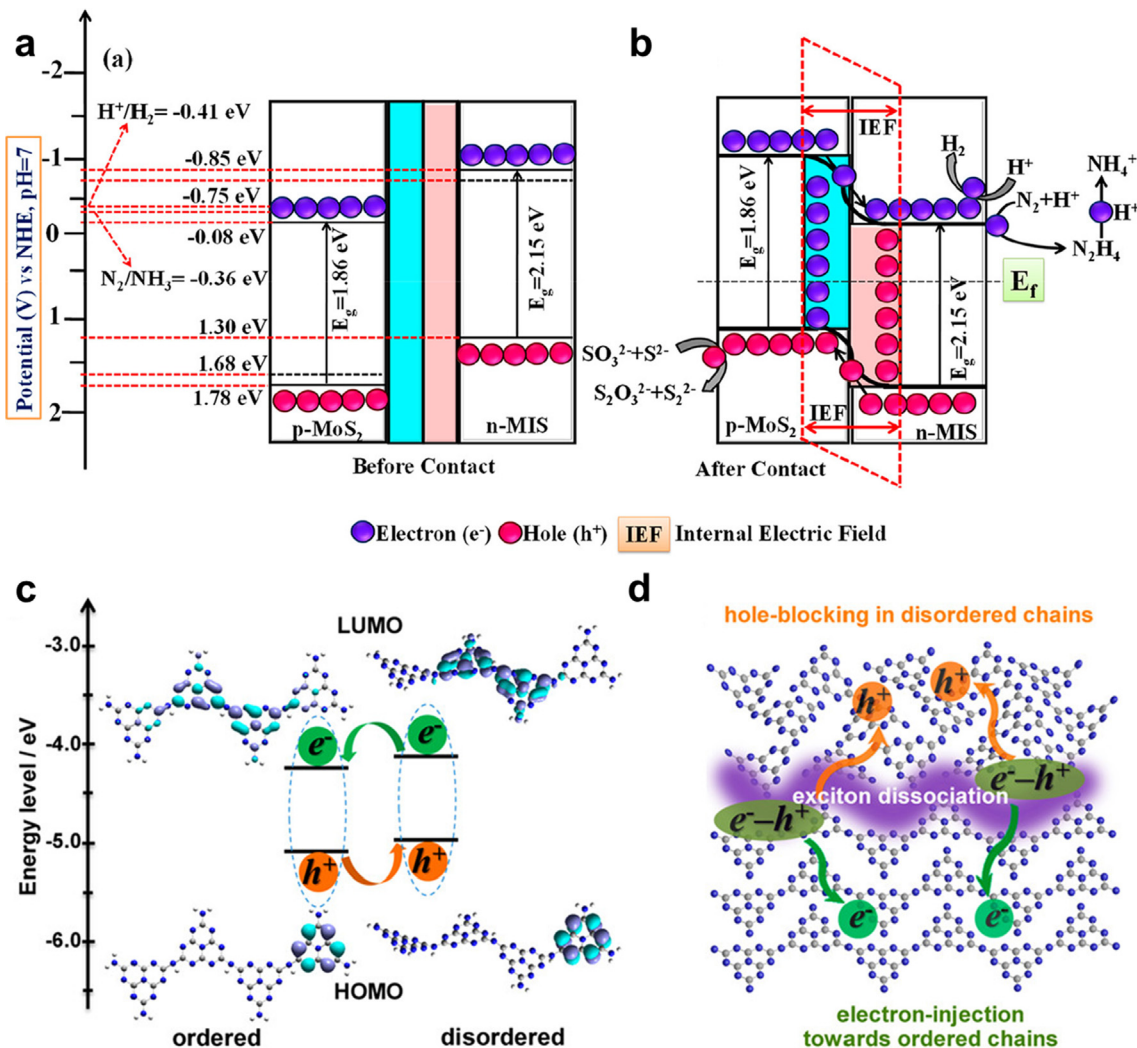


Fig. 5. Schematic illustration of band gap as well as band edge potential of neat MoS₂, neat MgIn₂S₄ (MIS) and MoS₂/MIS heterojunction photocatalyst (a) before contact and (b) after contact. Reproduced with permission [109]. Copyright 2020, American Chemical Society. (c) DFT simulations of ordered and disordered HM chains. Inset: corresponding charge-carrier transfer behaviors (blue balls, nitrogen atoms; gray balls, carbon atoms). Reproduced with permission [74]. Copyright 2017, American Chemical Society. (d) Schematic illustration of exciton dissociation and charge transfer in SC-HM. Reproduced with permission [74]. Copyright 2017, American Chemical Society.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors acknowledge financial support for this work from the Strategic Priority Research Program of CAS (XDB36000000), the National Natural Science Foundation of China (21422303, 21573049, 21872043, 22002028), National Basic Research Plan of China (2016YFA0201600), Beijing Natural Science Foundation (2142036), Youth Innovation Promotion Association, and Special Program of "One Belt One Road" of CAS.

References

- [1] X. Chen, S. Shen, L. Guo, S.S. Mao, Semiconductor-based photocatalytic hydrogen generation, *Chem. Rev.* 110 (2010) 6503–6570.
- [2] T. Hisatomi, J. Kubota, K. Domen, Recent advances in semiconductors for photocatalytic and photoelectrochemical water splitting, *Chem. Soc. Rev.* 43 (2014) 7520–7535.
- [3] A. Mills, S. Le Hunte, An overview of semiconductor photocatalysis, *J. Photochem. Photobiol. A: Chem.* 108 (1997) 1–35.
- [4] M.G. Walter, E.L. Warren, J.R. McKone, S.W. Boettcher, Q. Mi, E.A. Santori, N.S. Lewis, Solar water splitting cells, *Chem. Rev.* 110 (2010) 6446–6473.
- [5] H. Wang, X. Zhang, Y. Xie, Photoresponsive polymeric carbon nitride-based materials: design and application, *Mater. Today* 23 (2019) 72–86.
- [6] M.Z. Rahman, C.B. Mullins, Understanding charge transport in carbon nitride for enhanced photocatalytic solar fuel production, *Acc. Chem. Res.* 52 (2019) 248–257.
- [7] S. Cao, J. Low, J. Yu, M. Jaroniec, Polymeric photocatalysts based on graphitic carbon nitride, *Adv. Mater.* 27 (2015) 2150–2176.
- [8] G. Zhang, Z.-A. Lan, X. Wang, Conjugated polymers: catalysts for photocatalytic hydrogen evolution, *Angew. Chem. Int. Ed.* 55 (2016) 15712–15727.
- [9] X. Wang, K. Maeda, A. Thomas, K. Takanabe, G. Xin, J.M. Carlsson, K. Domen, M. Antonietti, A metal-free polymeric photocatalyst for hydrogen production from water under visible light, *Nat. Mater.* 8 (2009) 76–80.

- [10] A.I. Hochbaum, P. Yang, Semiconductor Nanowires for Energy Conversion, *Chem. Rev.* 110 (2010) 527–546.
- [11] P. Chen, Y. Tong, C. Wu, Y. Xie, Surface/Interfacial engineering of inorganic low-dimensional electrode materials for electrocatalysis, *Acc. Chem. Res.* 51 (2018) 2857–2866.
- [12] Q. Li, T. Lian, Exciton dissociation dynamics and light-driven H₂ generation in colloidal 2D cadmium chalcogenide nanoplatelet heterostructures, *Nano Res.* 11 (2018) 3031–3049.
- [13] H. Wang, X. Zhang, Y. Xie, Recent progress in ultrathin two-dimensional semiconductors for photocatalysis, *Mater. Sci. Eng., R* 130 (2018) 1–39.
- [14] T. Le Bahers, K. Takanebe, Combined theoretical and experimental characterizations of semiconductors for photoelectrocatalytic applications, *J. Photochem. Photobiol. C: Photochem. Rev.* 40 (2019) 212–233.
- [15] N.C. Giebink, G.P. Wiederrecht, M.R. Wasielewski, S.R. Forrest, Thermodynamic efficiency limit of excitonic solar cells, *Phys. Rev. B* 83 (2011) 195326.
- [16] K. Feron, J.W. Belcher, J.C. Fell, C.P. Dastoor, Organic solar cells: understanding the role of Förster resonance energy transfer, *Int. J. Mol. Sci.* 13 (2012) 17019–17047.
- [17] S. Melissen, T. Le Bahers, S.N. Steinmann, P. Sautet, Relationship between carbon nitride structure and exciton binding energies: a DFT perspective, *J. Phys. Chem. C* 119 (2015) 25188–25196.
- [18] K. Takanebe, Photocatalytic water splitting: quantitative approaches toward photocatalyst by design, *ACS Catal.* 7 (2017) 8006–8022.
- [19] O.V. Mikhnenko, P.W.M. Blom, T.-Q. Nguyen, Exciton diffusion in organic semiconductors, *Energy Environ. Sci.* 8 (2015) 1867–1888.
- [20] J. Pei, J. Yang, T. Yildirim, H. Zhang, Y. Lu, Many-body complexes in 2D semiconductors, *Adv. Mater.* 31 (2019) 1706945.
- [21] A. Köhler, H. Bässler, What controls triplet exciton transfer in organic semiconductors?, *J. Mater. Chem.* 21 (2011) 4003–4011.
- [22] T.M. Clarke, J.R. Durrant, Charge photogeneration in organic solar cells, *Chem. Rev.* 110 (2010) 6736–6767.
- [23] R.J. Holmes, S. Kéna-Cohen, V.M. Menon, S.R. Forrest, Strong coupling and hybridization of Frenkel and Wannier-Mott excitons in an organic-inorganic optical microcavity, *Phys. Rev. B* 74 (2006) 235211.
- [24] H.P. Hjalmarson, H. Büttner, J.D. Dow, Theory of core excitons, *Phys. Rev. B* 24 (1981) 6010–6019.
- [25] Z. Li, J. Feng, S. Yan, Z. Zou, Solar fuel production: Strategies and new opportunities with nanostructures, *Nano Today* 10 (2015) 468–486.
- [26] P.M. Rao, L. Cai, C. Liu, I.S. Cho, C.H. Lee, J.M. Weisse, P. Yang, X. Zheng, Simultaneously efficient light absorption and charge separation in WO₃/BiVO₄ core/shell nanowire photoanode for photoelectrochemical water oxidation, *Nano Lett.* 14 (2014) 1099–1105.
- [27] H. Wang, S. Jiang, S. Chen, D. Li, X. Zhang, W. Shao, X. Sun, J. Xie, Z. Zhao, Q. Zhang, Y. Tian, Y. Xie, Enhanced singlet oxygen generation in oxidized graphitic carbon nitride for organic synthesis, *Adv. Mater.* 28 (2016) 6940–6945.
- [28] R. Chen, Y. Tang, Y. Wan, T. Chen, C. Zheng, Y. Qi, Y. Cheng, W. Huang, Promoting singlet/triplet exciton transformation in organic optoelectronic molecules: role of excited state transition configuration, *Sci. Rep.* 7 (2017) 6225.
- [29] J.-L. Brédas, D. Beljonne, V. Coropceanu, J. Cornil, Charge-transfer and energy-transfer processes in π -conjugated oligomers and polymers: a molecular picture, *Chem. Rev.* 104 (2004) 4971–5004.
- [30] S. Bai, N. Zhang, C. Gao, Y. Xiong, Defect engineering in photocatalytic materials, *Nano Energy* 53 (2018) 296–336.
- [31] H. Wang, X. Zhang, Y. Xie, Photocatalysis in two-dimensional black phosphorus: the roles of many-body effects, *ACS Nano* 12 (2018) 9648–9653.
- [32] L. Yuan, L. Huang, Exciton dynamics and annihilation in WS₂ 2D semiconductors, *Nanoscale* 7 (2015) 7402–7408.
- [33] R. Berera, R. van Grondelle, J.T.M. Kennis, Ultrafast transient absorption spectroscopy: principles and application to photosynthetic systems, *Photosynth. Res.* 101 (2009) 105–118.
- [34] R.J. Ellingson, M.C. Beard, J.C. Johnson, P. Yu, O.I. Micic, A.J. Nozik, A. Shabaev, A.L. Efros, Highly efficient multiple exciton generation in colloidal PbSe and PbS quantum dots, *Nano Lett.* 5 (2005) 865–871.
- [35] K. Wu, Q. Li, Y. Du, Z. Chen, T. Lian, Ultrafast exciton quenching by energy and electron transfer in colloidal CdSe nanosheet-Pt heterostructures, *Chem. Sci.* 6 (2015) 1049–1054.
- [36] D. Sun, Y. Rao, G.A. Reider, G. Chen, Y. You, L. Brézin, A.R. Harutyunyan, T.F. Heinz, Observation of rapid exciton-exciton annihilation in monolayer molybdenum disulfide, *Nano Lett.* 14 (2014) 5625–5629.
- [37] C. Ruckebusch, M. Sliwa, P. Pernot, A. de Juan, R. Tauler, Comprehensive data analysis of femtosecond transient absorption spectra: a review, *J. Photochem. Photobiol. C: Photochem. Rev.* 13 (2012) 1–27.
- [38] R.K. Chowdhury, S. Nandy, S. Bhattacharya, M. Karmakar, S.N.B. Bhaktha, P.K. Datta, A. Taraphder, S.K. Ray, Ultrafast time-resolved investigations of excitons and biexcitons at room temperature in layered WS₂, *2D Mater.* 6 (2019) 015011.
- [39] J. Guo, H. Ohkita, H. Bente, S. Ito, Near-IR femtosecond transient absorption spectroscopy of ultrafast polaron and triplet exciton formation in polythiophene films with different regioregularities, *J. Am. Chem. Soc.* 131 (2009) 16869–16880.
- [40] B.K. Meyer, H. Alves, D.M. Hofmann, W. Kriegseis, D. Forster, F. Bertram, J. Christen, A. Hoffmann, M. Straßburg, M. Dworak, U. Haboeck, A.V. Rodina, Bound exciton and donor-acceptor pair recombinations in ZnO, *Phys. Stat. Sol. (b)* 241 (2004) 231–260.
- [41] J.B. Schlager, K.A. Bertness, P.T. Blanchard, L.H. Robins, A. Roshko, N.A. Sanford, Steady-state and time-resolved photoluminescence from relaxed and strained GaN nanowires grown by catalyst-free molecular-beam epitaxy, *J. Appl. Phys.* 103 (2008) 124309.
- [42] A. Splendiani, L. Sun, Y. Zhang, T. Li, J. Kim, C.-Y. Chim, G. Galli, F. Wang, Emerging photoluminescence in monolayer MoS₂, *Nano Lett.* 10 (2010) 1271–1275.
- [43] H. Zhang, Y. Chen, R. Lu, R. Li, A. Yu, Charge carrier kinetics of carbon nitride colloid: a femtosecond transient absorption spectroscopy study, *PCCP* 18 (2016) 14904–14910.
- [44] M.C. Beard, Multiple exciton generation in semiconductor quantum dots, *J. Phys. Chem. Lett.* 2 (2011) 1282–1288.
- [45] M.C. Beard, R.J. Ellingson, Multiple exciton generation in semiconductor nanocrystals: toward efficient solar energy conversion, *Laser Photonics Rev.* 2 (2008) 377–399.
- [46] J.A. McGuire, J. Joo, J.M. Pietryga, R.D. Schaller, V.I. Klimov, New aspects of carrier multiplication in semiconductor nanocrystals, *Acc. Chem. Res.* 41 (2008) 1810–1819.
- [47] M.C. Beard, K.P. Knutsen, P. Yu, J.M. Luther, Q. Song, W.K. Metzger, R.J. Ellingson, A.J. Nozik, Multiple Exciton Generation in Colloidal Silicon Nanocrystals, *Nano Lett.* 7 (2007) 2506–2512.
- [48] H. Zhu, Z. Chen, K. Wu, T. Lian, Wavelength dependent efficient photoreduction of redox mediators using type II ZnSe/CdS nanorod heterostructures, *Chem. Sci.* 5 (2014) 3905–3914.
- [49] H. Wang, D. Yong, S. Chen, S. Jiang, X. Zhang, W. Shao, Q. Zhang, W. Yan, B. Pan, Y. Xie, Oxygen-vacancy-mediated exciton dissociation in BiOBr for boosting charge-carrier-involved molecular oxygen activation, *J. Am. Chem. Soc.* 140 (2018) 1760–1766.
- [50] R. Godin, Y. Wang, M.A. Zwijnenburg, J. Tang, J.R. Durrant, Time-resolved spectroscopic investigation of charge trapping in carbon nitrides photocatalysts for hydrogen generation, *J. Am. Chem. Soc.* 139 (2017) 5216–5224.
- [51] H. Zhu, N. Song, W. Rodríguez-Córdoba, T. Lian, Wave function engineering for efficient extraction of up to nineteen electrons from one CdSe/CdS quasi-type II quantum dot, *J. Am. Chem. Soc.* 134 (2012) 4250–4257.
- [52] K. Wu, H. Zhu, Z. Liu, W. Rodríguez-Córdoba, T. Lian, Ultrafast charge separation and long-lived charge separated state in photocatalytic CdS-Pt nanorod heterostructures, *J. Am. Chem. Soc.* 134 (2012) 10337–10340.
- [53] Q. Li, F. Zhao, C. Qu, Q. Shang, Z. Xu, L. Yu, J.R. McBride, T. Lian, Two-dimensional morphology enhances light-driven H₂ generation efficiency in CdS nanoplatelet-Pt heterostructures, *J. Am. Chem. Soc.* 140 (2018) 11726–11734.
- [54] H. Zhu, T. Lian, Enhanced multiple exciton dissociation from CdSe quantum rods: the effect of nanocrystal shape, *J. Am. Chem. Soc.* 134 (2012) 11289–11297.
- [55] P.D. Cunningham, K.M. McCreary, B.T. Jonker, Auger recombination in chemical vapor deposition-grown monolayer WS₂, *J. Phys. Chem. Lett.* 7 (2016) 5242–5246.
- [56] N. Kumar, Q. Cui, F. Ceballos, D. He, Y. Wang, H. Zhao, Exciton-exciton annihilation in MoSe₂ monolayers, *Phys. Rev. B* 89 (2014) 125427.
- [57] F. Ma, Dynamics and coherent control of exciton-exciton annihilation in aqueous J-aggregate, *J. Phys. Chem. B* 122 (2018) 10746–10753.
- [58] H. Wang, S. Jiang, S. Chen, X. Zhang, W. Shao, X. Sun, Z. Zhao, Q. Zhang, Y. Luo, Y. Xie, Insights into the excitonic processes in polymeric photocatalysts, *Chem. Sci.* 8 (2017) 4087–4092.
- [59] Z. Zeng, X. Quan, H. Yu, S. Chen, Y. Zhang, H. Zhao, S. Zhang, Carbon nitride with electron storage property: enhanced exciton dissociation for high-efficient photocatalysis, *Appl. Catal. B Environ.* 236 (2018) 99–106.
- [60] K. Wu, A. Bera, C. Ma, Y. Du, Y. Yang, L. Li, T. Wu, Temperature-dependent excitonic photoluminescence of hybrid organometal halide perovskite films, *PCCP* 16 (2014) 22476–22481.
- [61] G. Wang, E. Palleau, T. Amand, S. Tongay, X. Marie, B. Urbaszek, Polarization and time-resolved photoluminescence spectroscopy of excitons in MoSe₂ monolayers, *Appl. Phys. Lett.* 106 (2015) 112101.
- [62] B. Choudhury, K.K. Paul, D. Sanyal, A. Hazarika, P.K. Giri, Evolution of nitrogen-related defects in graphitic carbon nitride nanosheets probed by positron annihilation and photoluminescence spectroscopy, *J. Phys. Chem. C* 122 (2018) 9209–9219.
- [63] M.S. Gaponenko, A.A. Lutich, N.A. Tolstik, A.A. Onushchenko, A.M. Malyarevich, E.P. Petrov, K.V. Yumashev, Temperature-dependent photoluminescence of PbS quantum dots in glass: evidence of exciton state splitting and carrier trapping, *Phys. Rev. B* 82 (2010) 125320.

- [64] C. Pu, H. Qin, Y. Gao, J. Zhou, P. Wang, X. Peng, Synthetic Control of Exciton Behavior in Colloidal Quantum Dots, *J. Am. Chem. Soc.* 139 (2017) 3302–3311.
- [65] D.J. Norris, A.L. Efros, M. Rosen, M.G. Bawendi, Size dependence of exciton fine structure in CdSe quantum dots, *Phys. Rev. B* 53 (1996) 16347–16354.
- [66] C. Ye, J.-X. Li, Z.-J. Li, X.-B. Li, X.-B. Fan, L.-P. Zhang, B. Chen, C.-H. Tung, L.-Z. Wu, Enhanced driving force and charge separation efficiency of protonated g-C₃N₄ for photocatalytic O₂ evolution, *ACS Catal.* 5 (2015) 6973–6979.
- [67] Y. Jiang, X. Wang, A. Pan, Properties of Excitons and Photogenerated Charge Carriers in Metal Halide Perovskites, *Adv. Mater.* 31 (2019) 1806671.
- [68] S.R. Scully, M.D. McGehee, Effects of optical interference and energy transfer on exciton diffusion length measurements in organic semiconductors, *J. Appl. Phys.* 100 (2006) 034907.
- [69] H. Gommans, S. Schols, A. Kadashchuk, P. Heremans, S.C.J. Meskers, Exciton diffusion length and lifetime in subphthalocyanine films, *J. Phys. Chem. C* 113 (2009) 2974–2979.
- [70] S.D. Stranks, G.E. Eperon, G. Grancini, C. Menelaou, M.J.P. Alcocer, T. Leijtens, L.M. Herz, A. Petrozza, H.J. Snaith, Electron-hole diffusion lengths exceeding 1 micrometer in an organometal trihalide perovskite absorber, *Science* 342 (2013) 341.
- [71] P.E. Shaw, A. Ruseckas, I.D.W. Samuel, Exciton diffusion measurements in poly(3-hexylthiophene), *Adv. Mater.* 20 (2008) 3516–3520.
- [72] R.R. Lunt, N.C. Giebink, A.A. Belak, J.B. Benziger, S.R. Forrest, Exciton diffusion lengths of organic semiconductor thin films measured by spectrally resolved photoluminescence quenching, *J. Appl. Phys.* 105 (2009) 053711.
- [73] S. Chen, W.M. Chen, I.A. Buyanova, Effects of strong band-tail states on exciton recombination dynamics in dilute nitride GaP/GaN core/shell nanowires, *J. Phys. Chem. C* 122 (2018) 19212–19218.
- [74] H. Wang, X. Sun, D. Li, X. Zhang, S. Chen, W. Shao, Y. Tian, Y. Xie, Boosting hot-electron generation: exciton dissociation at the order-disorder interfaces in polymeric photocatalysts, *J. Am. Chem. Soc.* 139 (2017) 2468–2473.
- [75] G.R. Hayes, I.D.W. Samuel, R.T. Phillips, Exciton dynamics in electroluminescent polymers studied by femtosecond time-resolved photoluminescence spectroscopy, *Phys. Rev. B* 52 (1995) R11569–R11572.
- [76] Y. Ren, Y. Dong, J.W.Y. Lam, B.Z. Tang, K.S. Wong, Studies on the aggregation-induced emission of silole film and crystal by time-resolved fluorescence technique, *Chem. Phys. Lett.* 402 (2005) 468–473.
- [77] J.-L. Wu, F.-C. Chen, Y.-S. Hsiao, F.-C. Chien, P. Chen, C.-H. Kuo, M.H. Huang, C.-S. Hsu, Surface plasmonic effects of metallic nanoparticles on the performance of polymer bulk heterojunction solar cells, *ACS Nano* 5 (2011) 959–967.
- [78] X. Li, S. Guo, C. Kan, J. Zhu, T. Tong, S. Ke, W.C.H. Choy, B. Wei, Au multimer@MoS₂ hybrid structures for efficient photocatalytic hydrogen production via strongly plasmonic coupling effect, *Nano Energy* 30 (2016) 549–558.
- [79] B. Miller, J. Lindlau, M. Bommer, A. Neumann, H. Yamaguchi, A. Holleitner, A. Högele, U. Wurstbauer, Tuning the Fröhlich exciton-phonon scattering in monolayer MoS₂, *Nat. Commun.* 10 (2019) 807.
- [80] Q. Zhang, J. Zhang, M.I.B. Utama, B. Peng, M. de la Mata, J. Arbiol, Q. Xiong, Exciton-phonon coupling in individual ZnTe nanorods studied by resonant Raman spectroscopy, *Phys. Rev. B* 85 (2012) 085418.
- [81] P. Kumar, E. Vahidzadeh, U.K. Thakur, P. Kar, K.M. Alam, A. Goswami, N. Mahdi, K. Cui, G.M. Bernard, V.K. Michaelis, K. Shankar, C₃N₄: a low bandgap semiconductor containing an azo-linked carbon nitride framework for photocatalytic, photovoltaic and adsorbent applications, *J. Am. Chem. Soc.* 141 (2019) 5415–5436.
- [82] T. Yuan, F. Yuan, X. Li, Y. Li, L. Fan, S. Yang, Fluorescence–phosphorescence dual emissive carbon nitride quantum dots show 25% white emission efficiency enabling single-component WLEDs, *Chem. Sci.* 10 (2019) 9801–9806.
- [83] L. Tian, W. Xie, X. Wu, B. Guo, G. Xie, P. Cheng, X. Liu, J.R. Gong, Overall regulation of exciton dynamics by defect engineering in polymeric photocatalysts for hydrogen evolution, *J. Phys. Chem. C* 124 (2020) 24667–24676.
- [84] X.-F. Shi, X.-Y. Xia, G.-W. Cui, N. Deng, Y.-Q. Zhao, L.-H. Zhuo, B. Tang, Multiple exciton generation application of PbS quantum dots in ZnO@PbS/graphene oxide for enhanced photocatalytic activity, *Appl. Catal. B Environ.* 163 (2015) 123–128.
- [85] Y.-Y. Li, B.-X. Zhou, H.-W. Zhang, S.-F. Ma, W.-Q. Huang, W. Peng, W. Hu, G.-F. Huang, Doping-induced enhancement of crystallinity in polymeric carbon nitride nanosheets to improve their visible-light photocatalytic activity, *Nanoscale* 11 (2019) 6876–6885.
- [86] N. Shehzad, M. Tahir, K. Johari, T. Murugesan, M. Hussain, A critical review on TiO₂ based photocatalytic CO₂ reduction system: strategies to improve efficiency, *J. CO₂ Util.* 26 (2018) 98–122.
- [87] Q. Zhang, D. Thrithamarassery Gangadharan, Y. Liu, Z. Xu, M. Chaker, D. Ma, Recent advancements in plasmon-enhanced visible light-driven water splitting, *J. Materiomics* 3 (2017) 33–50.
- [88] Z. Liu, W. Hou, P. Pavaskar, M. Aykol, S.B. Cronin, Plasmon resonant enhancement of photocatalytic water splitting under visible illumination, *Nano Lett.* 11 (2011) 1111–1116.
- [89] A.J. Nozik, Making the most of photons, *Nat. Nanotechnol.* 4 (2009) 548–549.
- [90] X. Huang, S. Han, W. Huang, X. Liu, Enhancing solar cell efficiency: the search for luminescent materials as spectral converters, *Chem. Soc. Rev.* 42 (2013) 173–201.
- [91] C. Smith, D. Binks, Multiple exciton generation in colloidal nanocrystals, *Nanomaterials* 4 (2014) 19–45.
- [92] J.J.H. Pijpers, R. Ulbricht, K.J. Tielrooij, A. Osherov, Y. Golan, C. Delerue, G. Allan, M. Bonn, Assessment of carrier-multiplication efficiency in bulk PbSe and PbS, *Nat. Phys.* 5 (2009) 811–814.
- [93] G. Nair, L.-Y. Chang, S.M. Geyer, M.G. Bawendi, Perspective on the prospects of a carrier multiplication nanocrystal solar cell, *Nano Lett.* 11 (2011) 2145–2151.
- [94] Y.-R. Lv, R. Huo, S.-Y. Yang, Y.-Q. Liu, X.-J. Li, Y.-H. Xu, Self-assembled synthesis of PbS quantum dots supported on polydopamine encapsulated BiVO₄ for enhanced visible-light-driven photocatalysis, *Sep. Purif. Technol.* 197 (2018) 281–288.
- [95] Y. Yan, R.W. Crisp, J. Gu, B.D. Chernomordik, G.F. Pach, A.R. Marshall, J.A. Turner, M.C. Beard, Multiple exciton generation for photoelectrochemical hydrogen evolution reactions with quantum yields exceeding 100%, *Nat. Energy* 2 (2017) 17052.
- [96] S. Ren, L.-Y. Chang, S.-K. Lim, J. Zhao, M. Smith, N. Zhao, V. Bulović, M. Bawendi, S. Gratečak, Inorganic-organic hybrid solar cell: bridging quantum dots to conjugated polymer nanowires, *Nano Lett.* 11 (2011) 3998–4002.
- [97] K. Wu, L.J. Hill, J. Chen, J.R. McBride, N.G. Pavlopolous, N.E. Richey, J. Pyun, T. Lian, Universal length dependence of rod-to-seed exciton localization efficiency in type I and quasi-type II CdSe/CdS nanorods, *ACS Nano* 9 (2015) 4591–4599.
- [98] M. Sim, J. Shin, C. Shim, M. Kim, S.B. Jo, J.-H. Kim, K. Cho, Dependence of exciton diffusion length on crystalline order in conjugated polymers, *J. Phys. Chem. C* 118 (2014) 760–766.
- [99] B. Guo, L. Tian, W. Xie, A. Batool, G. Xie, Q. Xiang, S.U. Jan, R. Boddula, J.R. Gong, Vertically aligned porous organic semiconductor nanorod array photoanodes for efficient charge utilization, *Nano Lett.* 18 (2018) 5954–5960.
- [100] A.T. Paradzah, K.D. Maabong, H.M.A.M. Elnour, A. Singh, M. Diale, T.P.J. Krüger, Identification of exciton-exciton annihilation in hematite thin films, *J. Phys. Chem. C* 123 (2019) 18676–18684.
- [101] S.F. Völker, A. Schmiedel, M. Holzapfel, K. Renziehausen, V. Engel, C. Lambert, Singlet-singlet exciton annihilation in an exciton-coupled squaraine-squaraine copolymer: a model toward hetero-J-aggregates, *J. Phys. Chem. C* 118 (2014) 17467–17482.
- [102] Z. Zhou, K. Li, W. Deng, J. Li, Y. Yan, Y. Li, X. Quan, T. Wang, Nitrogen vacancy mediated exciton dissociation in carbon nitride nanosheets: enhanced hydroxyl radicals generation for efficient photocatalytic degradation of organic pollutants, *J. Hazard. Mater.* 387 (2020) 122023.
- [103] E. Hendry, J.M. Schins, L.P. Candeias, L.D.A. Siebbeles, M. Bonn, Efficiency of exciton and charge carrier photogeneration in a semiconducting polymer, *Phys. Rev. Lett.* 92 (2004) 196601.
- [104] G. Zhang, G. Li, Z.-A. Lan, L. Lin, A. Savateev, T. Heil, S. Zafeirotos, X. Wang, M. Antonietti, Optimizing optical absorption, exciton dissociation, and charge transfer of a polymeric carbon nitride with ultrahigh solar hydrogen production activity, *Angew. Chem. Int. Ed.* 56 (2017) 13445–13449.
- [105] J. Yuan, Y. Tang, X. Yi, C. Liu, C. Li, Y. Zeng, S. Luo, Crystallization, cyanamide defect and ion induction of carbon nitride: Exciton polarization dissociation, charge transfer and surface electron density for enhanced hydrogen evolution, *Appl. Catal. B Environ.* 251 (2019) 206–212.
- [106] Y.-Y. Li, Y. Si, B.-X. Zhou, W.-Q. Huang, W. Hu, A. Pan, X. Fan, G.-F. Huang, Strategy to boost catalytic activity of polymeric carbon nitride: synergistic effect of controllable in situ surface engineering and morphology, *Nanoscale* 11 (2019) 16393–16405.
- [107] Z. Ma, P. Li, L. Ye, Y. Zhou, F. Su, C. Ding, H. Xie, Y. Bai, P.K. Wong, Oxygen vacancies induced exciton dissociation of flexible BiOCl nanosheets for effective photocatalytic CO₂ conversion, *J. Mater. Chem. A* 5 (2017) 24995–25004.
- [108] H. Sun, K. Wei, D. Wu, Z. Jiang, H. Zhao, T. Wang, Q. Zhang, P.K. Wong, Structure defects promoted exciton dissociation and carrier separation for enhancing photocatalytic hydrogen evolution, *Appl. Catal. B Environ.* 264 (2020) 118480.
- [109] G. Swain, S. Sultana, K. Parida, Constructing a novel surfactant-free MoS₂ nanosheet modified MgIn₂S₄ marigold microflower: an efficient visible-light driven 2D–2D p-n heterojunction photocatalyst toward HER and pH regulated NRR, *ACS Sustain. Chem. Eng.* 8 (2020) 4848–4862.
- [110] K.L. Corp, C.W. Schlenker, Ultrafast spectroscopy reveals electron-transfer cascade that improves hydrogen evolution with carbon nitride photocatalysts, *J. Am. Chem. Soc.* 139 (2017) 7904–7912.

- [111] S. Dong, C. Liu, Y. Chen, Boosting exciton dissociation and molecular oxygen activation by in-plane grafting nitrogen-doped carbon nanosheets to graphitic carbon nitride for enhanced photocatalytic performance, *J. Colloid Interface Sci.* 553 (2019) 59–70.
- [112] Y. Yang, C. Zhang, D. Huang, G. Zeng, J. Huang, C. Lai, C. Zhou, W. Wang, H. Guo, W. Xue, R. Deng, M. Cheng, W. Xiong, Boron nitride quantum dots decorated ultrathin porous g-C₃N₄: intensified exciton dissociation and charge transfer for promoting visible-light-driven molecular oxygen activation, *Appl. Catal. B Environ.* 245 (2019) 87–99.
- [113] J.S. Jang, H.G. Kim, J.S. Lee, Heterojunction semiconductors: a strategy to develop efficient photocatalytic materials for visible light water splitting, *Catal. Today* 185 (2012) 270–277.
- [114] H. Ou, X. Chen, L. Lin, Y. Fang, X. Wang, Biomimetic donor-acceptor motifs in conjugated polymers for promoting exciton splitting and charge separation, *Angew. Chem. Int. Ed.* 57 (2018) 8729–8733.
- [115] Z. Fang, S. Weng, X. Ye, W. Feng, Z. Zheng, M. Lu, S. Lin, X. Fu, P. Liu, Defect engineering and phase junction architecture of wide-bandgap ZnS for conflicting visible light activity in photocatalytic H₂ evolution, *ACS Appl. Mater. Interfaces* 7 (2015) 13915–13924.
- [116] F.A.L. Laskowski, M.R. Nellist, J. Qiu, S.W. Boettcher, Metal oxide/(oxy)hydroxide overlayers as hole collectors and oxygen-evolution catalysts on water-splitting photoanodes, *J. Am. Chem. Soc.* 141 (2019) 1394–1405.
- [117] Q.-Q. Zhou, Y.-Q. Zou, L.-Q. Lu, W.-J. Xiao, Visible-light-induced organic photochemical reactions through energy-transfer pathways, *Angew. Chem. Int. Ed.* 58 (2019) 1586–1604.
- [118] H. Wang, W. Liu, X. He, P. Zhang, X. Zhang, Y. Xie, An excitonic perspective on low-dimensional semiconductors for photocatalysis, *J. Am. Chem. Soc.* 142 (2020) 14007–14022.
- [119] H. Wang, S. Jin, X. Zhang, Y. Xie, Excitonic effects in polymeric photocatalysts, *Angew. Chem. Int. Ed.* 59 (2020) 22828–22839.